

Methods	$N_l = 3$	$N_l = 6$	$N_l = 12$	$N_l = 24$
Baseline	0.15 ± 0.07	0.27 ± 0.08	0.35 ± 0.06	0.49 ± 0.05
IIC (?)	0.22 ± 0.09	0.32 ± 0.07	0.41 ± 0.07	0.53 ± 0.06
Perone and Cohen-Adad (2018)	0.21 ± 0.09	0.31 ± 0.10	0.39 ± 0.07	0.50 ± 0.08
MLDS (?)	0.16 ± 0.15	0.35 ± 0.11	0.54 ± 0.09	0.59 ± 0.07
RPG (Ours)	0.21 ± 0.10	0.30 ± 0.08	0.45 ± 0.08	0.54 ± 0.08
RPG ⁺ (Ours)	0.31 ± 0.11	0.45 ± 0.10	0.55 ± 0.08	0.59 ± 0.08
Full Access ($N_l = 415$)	0.62 ± 0.05			

Table 5: Performance comparison on Retouch. Bold and underlined denote best and second best performance respectively.

erable improvements for the classes with more annotated pixels, while the performance for the clavicles only slightly improves. *RPG* also improves over the baseline for heart and lungs, but shows significant improvements for the difficult clavicles. Furthermore, *RPG*⁺ combines the aspects of *RPG* and augmentation-based consistency regularization, which noticeably improves all categories apart from the *right clavicle* with gains up to absolute 26% over the baseline. We display segmentation predictions in Fig. 3, where class-wise shortcomings of the different methods become visible.

Results on Extended JSRT. In the bottom half of Table 4, we display the results when using our fine-grained annotations of JSRT. The baseline of training simply on the annotated images achieves 10.43% mIoU. Both prediction-based pseudolabeling methods struggle in this complex low data environment and perform worse than the baseline. In contrast, *RPG* manages to correctly predict classes of the super-categories *Left Lung*, *Right Lung* and *Heart* leading to a mIoU of 31.53% thus improving upon the baseline by absolute 21.10%. *RPG*⁺ slightly boosts this further to a mIoU of 35.31%. We display the extended anatomy segmentations in Fig. 4. We see *RPG*⁺ managing to reconstruct the lung-subcategories, ventricles of the heart and the sub-diaphragm, but struggling with explicit predictions of bone structures.

Results on Spectralis. We display our results for the Spectralis dataset in Table 5. Here, we also see *RPG* outperforming the baseline for all considered N_l , thus showing its usability for the multi-class segmentation setting. *RPG*⁺ noticeably outperforms other methods especially for the low data schemes of $N_l = 3$ and $N_l = 6$ by up to 15% while having the same performance as MLDS for $N_l = 24$. We

display qualitative comparisons in Fig. 5.

Conclusion

In this work, we proposed a novel way of generating supervision for segmentation. We use labeled images as reference material, match pixels in an unlabeled image to their semantic counterparts, and allocate the corresponding label seen in the reference. This way, we do not fall into pitfalls common with prediction-based pseudo-labeling such as confirmation bias. Since no additional networks or alterations to a given architecture are necessary, our proposed method can easily be plugged into any existing framework. We argue that this way of pseudo-label generation is especially fitting for medical image analysis due structural similarity provided by underlying anatomical structures. We demonstrate the effectiveness of our approach through extensive experiments on chest X-ray anatomy segmentation and retinal fluid segmentation. We achieve fully supervised performance with only a handful of samples, thus, cutting the annotation cost by 95%. Our code and additional information are available in the supplementary.

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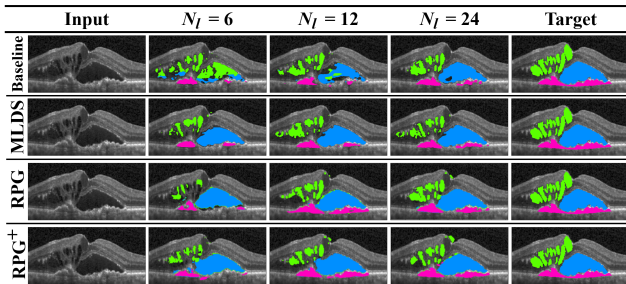


Figure 5: Qualitative Results on RETOUCH.

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